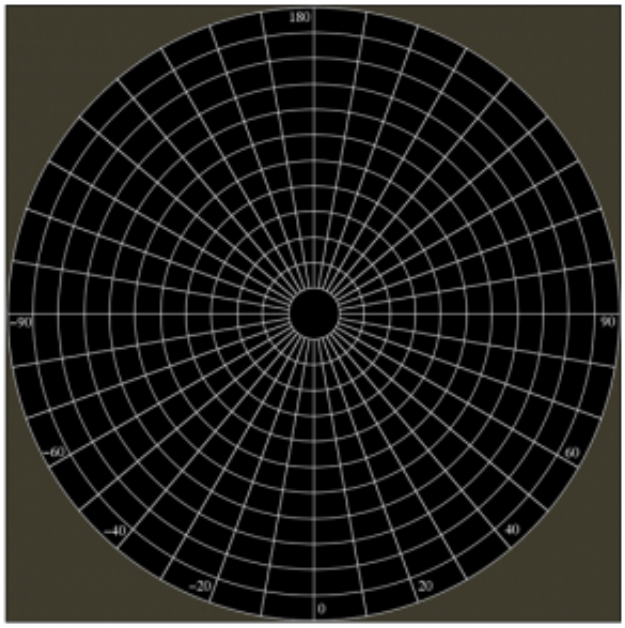
BusseLab Wiki

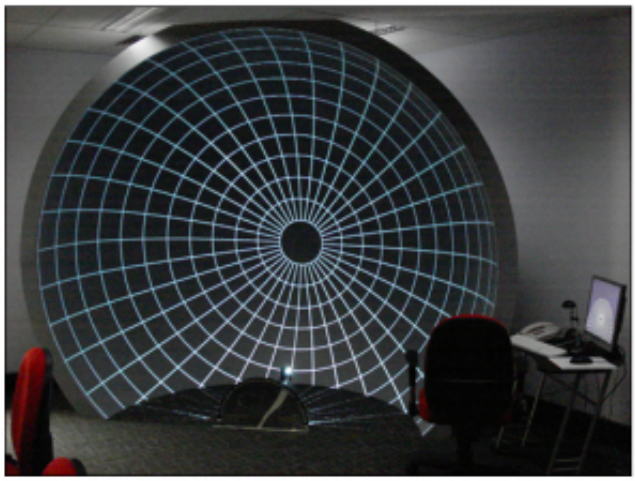
Dome projection

To use a standard projector to project a stimulus onto the dome surface, a spherical mirror is needed. But the mirror introduces a distortion that has to be corrected by applying a warping map to every frame. Some explanations and introduction of terms: http://paulbourke.net/dataformats/meshwarp/ [http://paulbourke.net/dataformats/meshwarp/]

Useful links:
http://paulbourke.net/dome/mirrordome/ [http://paulbourke.net/dome/mirrordome/]
http://paulbourke.net/dome/meshmapper/ [http://paulbourke.net/dome/meshmapper/]

The basic idea for finding the correct wrap is that one inputs an image for which the result on the projection surface is known. For example, in a dome, in a dome one knows the correct appearance of a polar grid (lines of longitude and latitude):

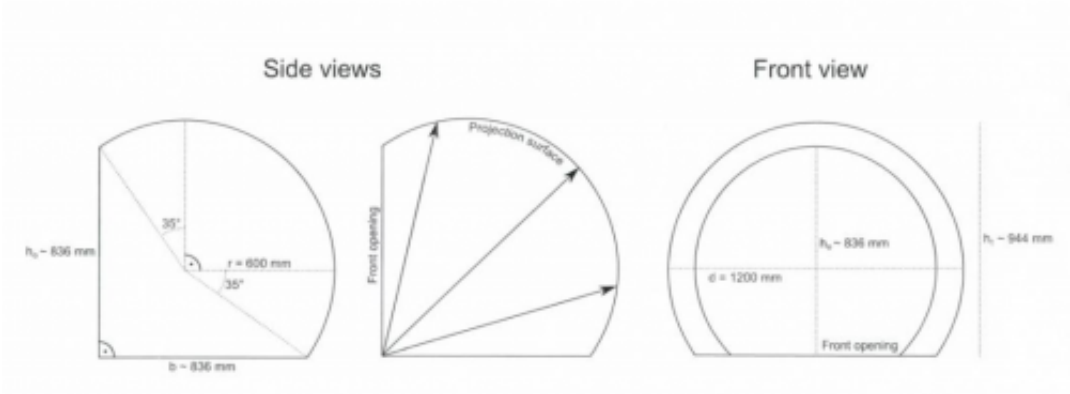




PhenoSys Dome

The dome setup, including dome, mirror and the frame that it is mounted on, is from PhenoSys https://www.phenosys.com/products/virtual-reality/jetball-dome/ [https://www.phenosys.com/products/virtual-reality/jetball-dome/].

The dimensions of the dome can be seen in this schematics:



The circumference is 377 cm.

This is the manual: phenosys_usermanual-jet_ball_dome_11_4.pdf

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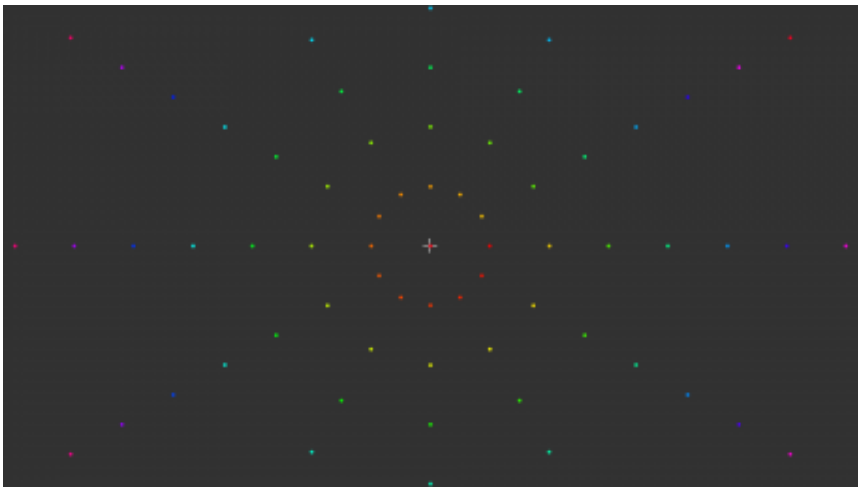
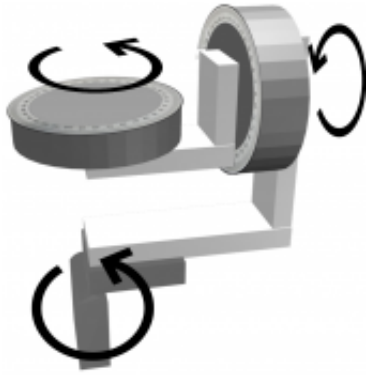
'Undistortion' with Psychtoolbox

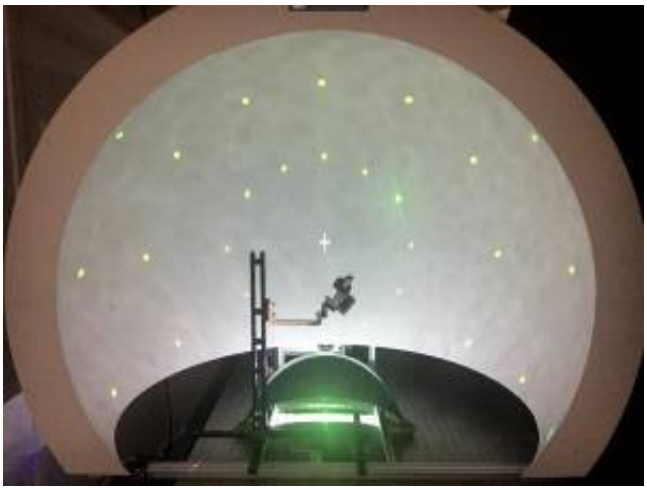
http://psychtoolbox.org/docs/DisplayUndistortionBVL [http://psychtoolbox.org/docs/DisplayUndistortionBVL]

This command projects a grid of dots that can be moved to align them correctly on the dome, this new locations are saved as a warping mesh, reflecting the setup-specific distortions.



A calibration device is used to indicate the correct positions of the single dots, in visual degree from the mouse's point of view. The device consist of a laser pointer that is adjustable in three spatial axes.





Steps in calibration code:

1. The command to start a calibration procedure is: [scal] = DisplayUndistortionBVL(caliboutfilename, screenid, shape, xnum, ynum, referenceImage, stereomode)
    - caliboutfilename: Path and filename
    - screenid: usually highest number of screens connected
    - shape: if 'circle', radial grid will be used, else square pattern, for the rest of the documentation it will be about radial pattern
    - x/ynum: only applies for square pattern, number of dots in x and y direction
    - referenceImage: path to an image in the background (only works when first initializing a file)
    - stereomode: if there are several displays to calibrate
2. If an existing calibration file has to be modifed,this command is used: [scal] = DisplayUndistortionBVL([caliboutfilename][, calibinfilename])
3. Due to the scale on the calibration device, only 15 or 30 degree distances between the dots are possible
4. scal.Nx/ydotsG and scal.X/YCALIBDOTS_ORG has to be commented or uncommented accordingly (around line 396 and 445 in the code)
5. Then the calibration starts
6. All dots that are not visible on the screen anymore will not be saved (especially for the lower part of the dome this is very helpful)

Use of the calibration file:

This file has to be used in the script the runs the visual stimulation with: PsychImaging('PrepareConfiguration'); PsychImaging('AddTask', 'AllViews', 'GeometryCorrection', 'path_to_file')

1. this function is called: [warpstruct, filterMode] = CreateDisplayWarp(window, calibfilename, showCalibOutput, varargin)
2. which calls this function: [xq,yq,vq] = griddata(varargin)
3. the parameters that are given to griddata are:
    - scal.X/YCALIBDOTS: original position of dots
    - scal.SELECTX/YCALIBDOTS: new position of dots
    - vertexCoords: vertexes produced by CreateDisplayWarp to divide screen equally
    - 'v4': Method of Interpolation
4. Interpolation Methods:
    - v4: uses Green's function and is default for BVL undistortion
    - cubic: uses triangle-based cubic interpolation
    - nearest: uses nearest neighbour interpolation with scatteredInterpolant

[https://de.mathworks.com/help/matlab/ref/scatteredinterpolant.html#bvkmyb8-Method] and adds as default extrapolation method nearest
- natural and linear: uses scatteredInterpolant [https://de.mathworks.com/help/matlab/ref/scatteredinterpolant.html#bvkmyb8-Method] without extrapolation (possible extrapolation: nearest and linear)

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